

Skills Summary

Choosing Expanded or Factored Form

Use this reference while completing your worksheet or when studying.

Skill 1 — Writing it in standard form

What expanded form shows: the constant term (the value at $x = 0$), the leading coefficient, and the degree — all readable off the page. **How:** every term of the first bracket meets every term of the second, then collect. $(x + p)(x + q) = x^2 + (p + q)x + pq$.

Example: Write $(x + 6)(x - 2)$ in standard form.

Step 1. The question asks for the constant term, which the factored form never shows on its own, so I expand.

Step 2. $x \cdot x = x^2$, $x \cdot (-2) = -2x$, $6 \cdot x = 6x$, $6 \cdot (-2) = -12$.

Step 3. Collect the middle: $-2x + 6x = 4x$.

$\Rightarrow x^2 + 4x - 12$, so the constant term is -12 .

Try it: Write $(x - 5)(x + 8)$ in standard form.

check: $x^2 - 3x - 40$

Skill 2 — Writing it in factored form

What factored form shows: what the expression is divisible by, and — once you set it equal to zero — where it is zero. **How, for $x^2 + bx + c$:** find two numbers with product c and sum b . Always take out a common factor first, and watch for a difference of squares: $x^2 - k^2 = (x - k)(x + k)$.

Example: Write $x^2 + 9x + 20$ in factored form.

Step 1. Nothing is common to both terms, and it is a trinomial with leading coefficient 1, so this is a product-and-sum search.

Step 2. Product 20, sum 9: the pair is 4 and 5, since $4 \cdot 5 = 20$ and $4 + 5 = 9$.

$\Rightarrow (x + 4)(x + 5)$

Try it: Write $x^2 - x - 12$ in factored form.

check: $(x + 4)(x - 3)$

Skill 3 — Letting the question choose the form

Zero-product property: if $AB = 0$ then $A = 0$ or $B = 0$. That is why zeros are a factored-form question and constant terms are an expanded-form question.

- Asked *where it is zero*, or *what it divides by*? Factor.
- Asked for the *constant term*, the *leading coefficient*, or a sum with another polynomial? Expand.

Example: Find the zeros of $x^2 - 2x - 8$.

Step 1. Zeros are asked for, so I want the form where a product equals zero — that means factoring first, not substituting values.

Step 2. Product -8 , sum -2 : the pair is -4 and 2 , giving $(x - 4)(x + 2)$.

Step 3. Set each factor to zero: $x - 4 = 0$ or $x + 2 = 0$.

$\Rightarrow (x - 4)(x + 2)$, so $x = 4$ or $x = -2$.

Try it: Find the zeros of $x^2 - 5x + 6$.

check: $x = 2$ or $x = 3$